

ISE5001: Technology Transfer and Commercialization
1st Semester 2026/2027

Fractus: Sell Products or Sell Technology?

(“Fractus”)

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Periodic reference will be made to this case (“Fractus”) by the professor during the course to illustrate various principles and themes in the course. It is essential that you read the case as part of your learning and assessment. This case will also be required for the individual assignment (Assessment Item 1), due on 8th November 2026.

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Case

Fractus:

Sell Products or Sell Technology?



In the summer of 2007, Rubén Bonet—a telecommunications engineer with an MBA—found himself facing a profound dilemma. The company he had co-founded in 1999 with Carles Puente, Fractus, was teetering on the edge of collapse. Just a year and a half earlier, during the Christmas holidays of 2005, they had felt invincible, convinced that Fractus was going to become huge. However, everything fell apart in just a few months.

A series of unexpected events had brought the company to the brink of the abyss. Early in the year, the company received an email from its Korean subsidiary reporting that LG, one of its major clients, had suddenly canceled 50% of its orders. Almost simultaneously, Vitelcom, their main client in Spain, went bankrupt. Then, in March 2006, Samsung, another of its key Korean clients in the mobile antenna manufacturing business, decided to cut ties with Fractus. In just six months, 80% of Fractus's hard-earned revenue had evaporated.

In Bonet's own words:

The aftermath was brutal! We were forced to make several layoffs, leaving the remaining team demoralized.

AUTHORSHIP CREDITS

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Yet, Fractus still possessed its unique fractal-based antenna technology. Surprisingly, it seemed that this original technology had spread across the mobile industry and had become dominant in the handheld device segment. However, Fractus had failed to monetize its intellectual property, leaving the company in a precarious position.

Bonet knew they had to act swiftly. During a crucial meeting with their investors, he proposed two initiatives: first, to explore how to legally defend their intellectual property rights (IPRs), and second, to devise a plan to revitalize their business. Both actions were urgent. For the latter, he foresaw two options: either to double down on the original OEM business of manufacturing and selling antennas, or to transform Fractus into a technology licensing company, leveraging their IPRs to generate revenue. This second path would be a very radical shift, but could it be the right move? Should they abandon their original manufacturing business entirely, or was there still a way to revive it? The stakes were high, and the path forward was fraught with uncertainty. Bonet and his team needed to carefully weigh their options and decide on the best course of action to save Fractus from the brink before the next board meeting in September 2007.

THE BEGINNINGS (1992–2000)

In 1992, Carles Puente graduated with a degree in Telecommunications Engineering from the Universitat Politècnica de Catalunya (UPC). Immediately afterward, he completed an MSc in Photonics at the University of Illinois (USA). In 1994, he returned to the UPC to begin a PhD in fractal¹ antennas. Puente's idea was highly innovative: Was it possible to develop communication antennas with fractal-based geometries? If so, these antennas would offer numerous advantages over existing ones: first, they would be multiband, since repeating geometrical shapes at multiple scales could produce the same radiation properties across different frequency bands; and second, by compressing long paths into a small area, they could significantly reduce antenna size. These two advantages would result in compact, multiband antennas, eliminating the annoying classic antenna rods that cell phones had at the time. For all these reasons, fractal-based antennas could be optimal for applications of mobile communications, wireless networks or radar systems, military, and aerospace applications.

As a result of his doctoral thesis, Carles Puente registered the first global patent^{11 1} for fractal antennas in 1995, at the Spanish Patent Office. They decided to patent solely at the national level due to its cost-effectiveness and streamlined process. Filing national patent applications would take 1 to 3 years and could cost between EUR 6,000 and EUR 10,000, while a European patent granting procedure would take 3 to 5 years and cost around EUR 20,000.^{2, 3}

¹ A fractal is a geometric figure that replicates itself at different scales. In nature, many forms resemble fractals, such as in the contours of coastlines, in the growth of trees from leaves to branches, or in snowflakes.

¹¹ A patent is an exclusive right granted for an invention—whether a product or a process—that generally provides a new way of doing something, or offers a new technical solution to a problem. To obtain a patent, technical information about the invention must be disclosed to the public through a patent application.

In 1998, foreseeing the commercial potential of fractal antenna technology, Puente contacted a former classmate, Rubén Bonet, who was then working in consulting, after completing an MBA. In that meeting, they outlined the basic ideas of what would become their future company: Fractus, which was incorporated in March 1999 as a spin-off from the UPC. Fractus was, in fact, one of the first technology-based startups to emerge from Barcelona's scientific ecosystem. With a clear business profile and orientation, Bonet assumed the role of Chief Executive Officer, while Puente continued as Chief Technology Officer. On the same day the company was founded, Bonet left his consulting job, and a month later, Puente left his position as a professor and researcher at UPC. The journey began with a capital of EUR 60,000 from the founders.

During the first few months of the company's existence, the first employees of the commercial and R&D teams came aboard. Bonet focused on developing the business plan and seeking investors, while Puente continued his research and worked on patenting his inventions. The first significant capital contribution came a few months later from a private investor based in Monaco: EUR 600,000 for 25% of the company, at a pre-money valuation of EUR 1.8 million. With this investment, Fractus financed a small antenna laboratory and commercial activities, including the development of its first website.

INITIAL BUSINESS DEPLOYMENT (2000–2003)

As a result of this commercial activity, the first meeting with one of the major telecommunications equipment manufacturers of the time, Ericsson, was held in Sweden in January 2000. The initial strategy focused on selling antennas for communication towers, and the product development process prioritized bringing to market those prototypes that were mature and ready for use. With the imminent deployment of 3G technology, Ericsson suggested the development of a new antenna tailored to this new generation of mobile communications, with the intention of integrating it into the equipment they had supplied to Telefónica. In June of that year, Fractus made its first sales: a set of small antennas delivered to RadioMobil, the leading Czech telecom operator at the time. Meanwhile, a burgeoning mobile communications system was taking shape, with Nokia at the forefront. Fractus's multiband, fractal-based antennas seemed ideally suited to support future generations of mobile communications while avoiding the cumbersome rod antennas of most mobile phones. However, contacts with Nokia were unsuccessful.

In early 2001, Fractus secured a EUR 600,000 contract with Telefónica to supply 2G/3G multiband antennas for their communication towers. Nevertheless, doubts among telecommunications operators about the functionality and business possibilities of 3G led to delays in its full deployment, thus jeopardizing Fractus's business model. In Bonet's words:

Fractus had received money from investors and many awards and recognitions from the press and Barcelona's technological environment, but it still lacked sustainable sales to support steady growth.

They needed funding and, if it did not come from customers, it should come from investors. Despite the uncertainty caused by the burst of the dot.com bubble in 2000, they managed to secure a EUR 2 million in new capital from the private equity firm 3i, at a post-money valuation of EUR 8 million, following an initial convertible loan of about EUR 450,000 from the public fund Barcelona Emprèn in January 2000 (see **Exhibit 1**). Given the increasing complexity of Fractus's financial management, a new finance director, Ángel García, joined the company in 2001.

By 2002, Fractus had achieved a turnover of EUR 1.3 million. However, the outlook remained bleak due to the persistent delays in 3G deployment by operators.

EXPANSION, DIVERSIFICATION, AND INTERNATIONALIZATION (2003–2006)

Several circumstances contributed to Fractus's success during the following years. First, they shifted their primary focus away from 2G/3G telecommunications towers by pivoting the business model towards manufacturing fractal-based antennas for mobile phones. Second, Carles Puente and his team had been working on the design of new multiband fractal-based antennas capable of supporting 2G, GPS, Bluetooth, or WiFi for mobile phones. Third, Nokia, the leader in mobile telephony at the time, had models with integrated antennas, but only with 1G capability. And fourth, they hired Michael Price, from Antetronic—a mobile antennas vendor—as the new Sales Manager, which significantly increased Fractus's commercial capacity in the emerging market of cell phone antennas.

Gradually, the doors of major mobile phone manufacturers opened to Fractus's products, with the company receiving orders ranging from 5,000 to 200,000 antennas. Vitelcom, a Málaga-based mobile phone manufacturer for Telefónica, became Fractus's main client. Manufacturing and selling antennas for mobile phones required multiple meetings with end customers to determine how to adapt the antenna to the design required by the customer. It was necessary to collaborate with concurrent engineering teams on customized projects for the new mobile phone models. This entailed a significant risk, as the company had to bear the R&D costs necessary to adapt the antennas to the required specifications. The antennas had to demonstrate certain features of mechanical resistance and radio-electric behavior within the context of the new mobile device to be finally approved. Sometimes, the client would cancel the project after months of development. Once a project was accepted, the production process involved working with an array of suppliers—such as plastic molders and metal part stampers—capable of guaranteeing precision manufacturing at high levels of quality (see **Exhibit 2**).

The future was promising. By the end of 2002, a new round of joint investment added EUR 8 million to the company's capital—EUR 2 million from 3i and EUR 6 million from Apax Partners—at a new post-money valuation of EUR 20 million. As a result, Bonet and Puente's shares were diluted, reducing their combined ownership to less than 50%.

Since 2002, Fractus had maintained an agreement with a South Korean distributor, who distributed antennas manufactured by Fractus to various subcontractors. In 2003, after market penetration and integration into the Korean manufacturing ecosystem—and with an eye on major clients like Samsung or LG—Fractus Korea Ltd was established. Competition in the Korean market was so fierce that it compelled Bonet to say:

Korean customers didn't care if one competitor copied another's design. All they wanted was the best antenna at the cheapest possible price.

By the summer of 2005, Fractus Korea had grown to a staff of over 20 employees. To better serve the Asian market, the company built a manufacturing plant in Yantai (China), located near South Korea and separated only by the Yellow Sea. Sales volumes were increasing significantly, driven by demand from LG, Samsung, and other Korean manufacturers, as well as European mobile antenna customers, with Vitelcom leading the way.

That same year, Fractus appointed an experienced executive, Torbjörn Folkebrant—a Swedish national and former board member of several NASDAQ-listed semiconductor companies—as Executive Chairman to project a more senior image and to help establish connections within large corporations as potential customers.

From 1999 to 2005, Fractus had successfully sold antennas, although it had not yet achieved profitability. (See P&L in **Exhibit 3** and Balance in **Exhibit 4** for financial details).

Suspicious of patent infringements in LG-related projects began to arise among Fractus's engineers, despite the strict confidentiality agreements signed by the manufacturers. Under U.S. law, anyone using patented technology could be declared guilty of patent infringement, even if unaware that the technology had a legal owner. Bonet and Puente believed that as long as Fractus held ownership of the key patents, they could compel non-client companies to pay royalties for the use of their technology in their devices. With this idea in mind, Fractus requested the approval of its patents from the United States Patent and Trademark Agency (USPTO) and, in July 2004, they traveled to Dallas to meet with McKenzie, a leading law firm specialized in intellectual and industrial property rights regulation. In August 2005, Fractus signed an agreement with McKenzie to legally protect its technology and to establish procedures to apply when a potential patent infringement from a large corporation was detected.

That same year, Fractus was recognized as a leading company specializing in the development and manufacturing of fractal-based antennas, highly diversified and increasingly globalized. In view of its ambitious and increasingly defined expansion plans, investors contributed a new capital of EUR 4.5 million, at a company post-money valuation of EUR 24.5 million.

CRISIS AND DOWNFALL (2006)

In early 2006, after a detailed analysis of more than 100 different designs from competitors and clients—in products not developed by Fractus—the company realized that almost all mobile phones with integrated antennas on the market were using technology originally patented by Fractus. This not only could constitute an infringement but also a serious limitation to the company's growth potential, as it enabled clients to access similar, cheaper designs, manufactured by suppliers who did not bear any R&D costs. Despite the troubling discovery, it was clear that Fractus's technology had reached success and had spread throughout the industry. The challenge now was to find a way to ensure that those using its technology paid for it. Even so, the possibilities for short-term action seemed limited, as the granting of patents by the USPTO in the U.S. still seemed to be distant.

Shortly after, bad news ensued. LG suddenly cut its contracts by 50% and transferred them to a competitor that appeared to be using Fractus's designs. Soon after, Vitelcom lost its contracts with Telefónica and went bankrupt due to a strategic shift that stemmed from a patent conflict with Nokia. Fractus's mobile business in Europe was also seriously hit. But this was not the end. Samsung also suspended its purchasing orders following another patent dispute, this time with another supplier from Boston, who accused Fractus of intellectual property infringement. In response, Samsung terminated all business with both companies. Faced with the dramatic drop in sales to Samsung and LG, Fractus decided to close its Korean subsidiary in an orderly manner. Within six months, the company's sales fell by 80%, forcing a radical three-phase restructuring plan that included laying off a significant part of their workforce.

Paradoxically, Fractus's engineers had taught clients and suppliers how to manufacture their antennas and the technology of fractal-based antennas was spreading throughout the mobile communications corporate world. Yet, for various reasons, everyone was surprisingly discontinuing purchases of Fractus's devices. In May 2006, Ron Epstein, an expert consultant in high-tech sectors, joined Fractus's board of directors to help the company understand the complex issues regarding intellectual property management.

A CRITICAL DECISION

On September 26, 2006, a critical board meeting was held, with a main item on the agenda: What to do? The only short-term survival option seemed to be finding a buyer interested in Fractus; mainly, in its technology protected by the patent portfolio. The company had enough cash to survive approximately one more year, during which time it would seek a buyer for 100% of its capital. For this sale, Fractus would be presented as a technology company with an exclusive and valuable intellectual property base, whose developments had become the dominant designs for mobile device antennas worldwide, even though it had yet to find a way to monetize this success.

To facilitate the company's sale, Fractus engaged Merger & Acquisition specialists in the USA. Two NASDAQ-listed companies showed interest: IntraDigikey, with an offer of USD 60 million,

and CoolCom, whose proposal was slightly higher at USD 65 million. IntraDigikey intended to integrate Fractus's patent portfolio into its own, with which it had synergies, while CoolCom sought to use the portfolio to strengthen its negotiating position in ongoing litigation against Nokia. Fractus's board decided to accept CoolCom's proposal. However, after several months of due diligence and acquisition preparations, the San Diego-based company decided to cancel the operation, citing issues related to the patents. Shortly thereafter, it was announced that CoolCom had reached an agreement with Nokia regarding their conflict over potential patent infringement. Bonet had a very clear idea of what had happened:

CoolCom had used Fractus's patents to strengthen its position in the negotiation with Nokia. Once they had signed the agreement with the Finnish company, CoolCom was no longer interested in Fractus.

Despite this new and serious setback, Fractus's confidence in the value of its intellectual property was reaffirmed. It was likely that CoolCom's lawyers and engineers, after analyzing Fractus's patents, had concluded that they were indeed valuable. The settlement agreement with Nokia likely led to the cancellation of the acquisition.

The process of selling the company continued, but further outreach to numerous financial agents in the U.S. proved to be unsuccessful. By early 2007, Fractus hit rock bottom. With only a couple of months of cash remaining, the board was legally obliged to consider voluntary insolvency proceedings—a step toward bankruptcy—to avoid personal legal liabilities. The main investors, Apax and 3i, were clearly seeking an exit. A decision had to be made within 30 days: either secure liquidity or close the company permanently.

On May 23, 2007—the final day of the legal deadline to secure investment or face bankruptcy—an investment of EUR 2 million arrived at the last moment from a Friends & Family pool, which included the public fund Barcelona Emprèn. Apax and 3i sold their shares, and Bonet assumed the role of chairman in addition to his position as CEO of the company. With the remaining shareholders divided over the path to follow, Bonet, as chairman, had to assume a mission: to save the remnants of the company and, within a year, to get it back on track.

The very first priority was clear: the company had to legally defend its intellectual property rights (IPR). The second challenge was much harder: how to revitalize the business. For those involved, there were only two viable alternatives in sight: either relaunch the antenna production and sales operation, with strict control over third-party licensing, or pivot entirely to a business model solely focused on research, development, and the licensing or sale of the company's intellectual property.

The first option of **manufacturing** and delivering the antennas themselves would provide continuity to what the company had been doing from the outset. It would likely allow them to capture a larger share of the value created, leading to higher profit margins. It would also allow greater control over quality, brand positioning, and customer experience by customizing the antennas, offering warranties, and maintaining direct relationships with the clients (smart device

manufacturers and network operators) through key account managers. However, this option would entail further investment in equipment and machinery, as well as additional time and costs for business development to secure projects.

Pivoting to the second option of **licensing** their patents would require the development of a completely new business, one in which Fractus had limited experience. This would require continuous technology surveillance of antenna and smart device manufacturers, and if any were to be found using Fractus's technology, the company would have to negotiate a licensing agreement (including royalties), which would lead to new legal expenditures: hiring IP litigation lawyers and consultants. This model would entail dismantling the antenna sales business: laying off employees and closing down laboratories and manufacturing facilities. However, to what extent would this model affect the company's operational costs, profit margins, and scalability? Would Fractus still be able to maintain control over final product quality or its brand?

Bonet commented to his co-founder, Puente:

We urgently need to choose one of those alternatives if we want to remain in business. In any case, we should carefully gauge and compare the risks and complexities—regulatory, operational, and commercial—of both options before our next board meeting.



Exhibits
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*Exhibit 1. Barcelona Emprèn and
Entrepreneurial Ecosystems*



Barcelona Emprèn was a venture capital firm created by public administrations to invest in technology or innovative startups, using a range of financial instruments and providing strategic support.⁴

Investment profile:

- ≈ Small companies with turnover under EUR 1.5 million
- ≈ Companies located in Barcelona or the greater Barcelona area
- ≈ Technology-based or innovative companies

Public entities invest in spin-offs because of their potential to generate positive externalities in the economy (growth, investment and talent attraction, quality employment, good salaries, among others). Good examples of this are the **entrepreneurial ecosystems**, such as those of MIT⁵ or Oxford (UK),⁶ which achieve significant entrepreneurial impact by focusing on real-world challenges and providing educational foundations in entrepreneurship, innovation, and intellectual property rights.

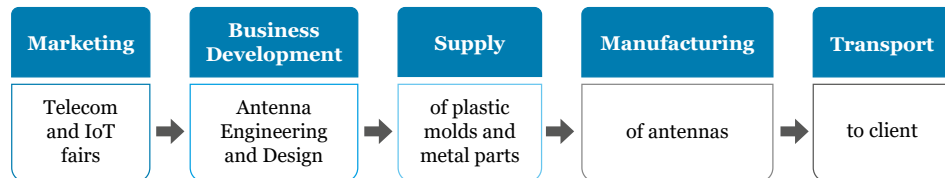
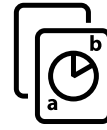
Due to the high level of risk associated with investing in spin-offs—as their market entry is neither linear nor predictable—it is difficult to attract investors. Therefore, **public instruments** are created to invest in the most embryonic phases of spin-offs, when they are still far from accessing the market. The purpose is to cover what is referred to in economic terms as “market failure”—the market’s reluctance to invest in R&D-intensive opportunities, even when this is economically and socially desirable—or what is described as the “Valley of Death” in entrepreneurial terminology: the transition between knowledge generated in a public research laboratory and the successful entry into the market in the form of a product accepted by consumers.

Source: Different sources, as indicated in the Exhibit’s text.



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Exhibit 2. *Value Chain of Fractus's
Antenna Sales for Mobile Phones*



Source: Prepared by the authors.

Exhibit 3. *Fractus P&L
Financial Statement from 1999
to 2006 (in thousands of EUR)*



Fractus	1999	2000	2001	2002	2003	2004	2005	2006	Total
Product & Services	55	358	477	1,282	1,311	3,316	3,787	2,240	12,828
Gross revenues	55	358	477	1,282	1,311	3,316	3,787	2,240	12,828
Product Cost		3	79	604	573	1,752	1,641	927	5,578
Net revenues	55	355	398	679	739	1,564	2,146	1,314	7,250
Staff	86	152		1,257	1,723	2,700	2,685	2,533	11,137
General Expenses	189	182	818	899	1,486	1,639	1,533	1,558	8,302
EBITDA	-220	22	-420	-1,477	-2,470	-2,775	-2,072	-2,777	-12,189
Depreciation	-12	-121	-86	-1,231	-237	-337	-750	-318	-3,092
Financials	-1	-5	-27	-32	31	29	-6	-2	-12
Extraordinary		6	199	5	300	369	1,362	306	2,547
Net profit	-233	-98	-334	-2,735	-2,376	-2,714	-1,466	-2,790	-12,746

Source: Data provided by the founders.



Exhibits
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*Exhibit 4. Fractus Balance
Sheet from 1999 to 2006
(in thousands of EUR)*



	1999	2000	2001	2002	2003	2004	2005	2006
<i>Fixed assets</i>	177	841	2,626	2,529	3,250	3,801	2,762	2,667
Intangible Assets	74	551	1,917	1,163	1,270	1,442	1,353	1,508
Property, Plant & Equipment	62	104	264	276	769	967	912	864
Long-term Investments	41	186	445	1,090	1,211	1,392	497	295
<i>Current assets</i>	368	318	1,500	5,333	6,064	4,113	2,198	3,215
Inventories	0	4	185	109	11	155	95	48
Receivables	208	122	577	407	486	771	1,030	660
Cash & Short-term Investments	160	192	738	4,817	5,567	3,187	1,073	2,507
<i>Assets & liabilities</i>	545	1,159	4,126	7,862	9,314	7,914	4,960	5,882
<i>Equity</i>	68	350	2,291	4,299	5,923	3,209	1,563	3,084
<i>Long term liabilities</i>	188	388	737	2,761	2,698	2,907	1,567	1,800
<i>Current liabilities</i>	289	421	1,098	802	693	1,798	1,830	998

Source: Data provided by the founders.

NOTES

- ¹ World Intellectual Property Organization, WIPO. (2024). Patents. Retrieved from: <https://www.wipo.int/patents/en/>
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